In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
df_csv = pd.read_csv("sales.csv")
print("CSV shape:", df_csv.shape)
df_csv.head()
```

```
CSV shape: (125, 6)
```

Out[2]:

| | OrderID | Product | Category | Quantity | Price | Region |
|---|---|---|---|---|---|---|
| 0 | 1 | Phone | Electronics | 2 | 1499.0 | South |
| 1 | 2 | Headphones | Accessories | 1 | 2499.0 | South |
| 2 | 3 | Laptop | Accessories | 2 | 1499.0 | South |
| 3 | 4 | Laptop | Accessories | 1 | 999.0 | North |
| 4 | 5 | Headphones | Accessories | 2 | 999.0 | East |

In [3]:
```
df_excel = pd.read_excel("sales.xlsx")
print("Excel shape:", df_excel.shape)
df_excel.head()
```

```
Excel shape: (125, 6)
```

Out[3]:

| | OrderID | Product | Category | Quantity | Price | Region |
|---|---|---|---|---|---|---|
| 0 | 1 | Phone | Electronics | 2 | 1499.0 | South |
| 1 | 2 | Headphones | Accessories | 1 | 2499.0 | South |
| 2 | 3 | Laptop | Accessories | 2 | 1499.0 | South |
| 3 | 4 | Laptop | Accessories | 1 | 999.0 | North |
| 4 | 5 | Headphones | Accessories | 2 | 999.0 | East |

In [4]:
```
df_json = pd.read_json("sales.json")
print("JSON shape:", df_json.shape)
df_json.head()
```

```
JSON shape: (125, 6)
```

Out[4]:

| | OrderID | Product | Category | Quantity | Price | Region |
|---|---|---|---|---|---|---|
| 0 | 1 | Phone | Electronics | 2 | 1499.0 | South |
| 1 | 2 | Headphones | Accessories | 1 | 2499.0 | South |
| 2 | 3 | Laptop | Accessories | 2 | 1499.0 | South |
| 3 | 4 | Laptop | Accessories | 1 | 999.0 | North |
| 4 | 5 | Headphones | Accessories | 2 | 999.0 | East |

In [5]:
```
df_csv.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 125 entries, 0 to 124
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   OrderID   125 non-null    int64
 1   Product   125 non-null    str
 2   Category  125 non-null    str
 3   Quantity  125 non-null    int64
 4   Price     117 non-null    float64
 5   Region    125 non-null    str
dtypes: float64(1), int64(2), str(3)
memory usage: 6.0 KB
```

In [6]:
```python
print("Missing values in each column:")
print(df_csv.isnull().sum())
print()
print("Duplicate rows:", df_csv.duplicated().sum())
```

```
Missing values in each column:
OrderID     0
Product     0
Category    0
Quantity    0
Price       8
Region      0
dtype: int64

Duplicate rows: 5
```

In [7]:
```python
def clean_data(df):
    df = df.copy()
    df["Price"] = df["Price"].fillna(df["Price"].median())
    df = df.drop_duplicates()
    return df

df_csv_clean = clean_data(df_csv)
print("Rows before cleaning:", len(df_csv))
print("Rows after cleaning:", len(df_csv_clean))
print("Missing values after cleaning:", df_csv_clean.isnull().sum().sum()
```

```
Rows before cleaning: 125
Rows after cleaning: 120
Missing values after cleaning: 0
```

In [8]:
```python
df_excel_clean = clean_data(df_excel)
df_json_clean = clean_data(df_json)

print("All three cleaned datasets are equal:", df_csv_clean.reset_index(d

df = df_csv_clean.reset_index(drop=True)
print("Unified dataframe shape:", df.shape)
df.head()
```

```
All three cleaned datasets are equal: True
Unified dataframe shape: (120, 6)
```

Out[8]:

| | OrderID | Product | Category | Quantity | Price | Region |
|---|---|---|---|---|---|---|
| 0 | 1 | Phone | Electronics | 2 | 1499.0 | South |
| 1 | 2 | Headphones | Accessories | 1 | 2499.0 | South |
| 2 | 3 | Laptop | Accessories | 2 | 1499.0 | South |
| 3 | 4 | Laptop | Accessories | 1 | 999.0 | North |
| 4 | 5 | Headphones | Accessories | 2 | 999.0 | East |

In [9]:
```
df["Total"] = df["Quantity"] * df["Price"]
df["PriceCategory"] = pd.cut(df["Price"], bins=[0, 1000, 2000, 3000], lab
df.head()
```

Out[9]:

| | OrderID | Product | Category | Quantity | Price | Region | Total | Pr |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | Phone | Electronics | 2 | 1499.0 | South | 2998.0 | |
| 1 | 2 | Headphones | Accessories | 1 | 2499.0 | South | 2499.0 | |
| 2 | 3 | Laptop | Accessories | 2 | 1499.0 | South | 2998.0 | |
| 3 | 4 | Laptop | Accessories | 1 | 999.0 | North | 999.0 | |
| 4 | 5 | Headphones | Accessories | 2 | 999.0 | East | 1998.0 | |

In [10]:
```
df.describe()
```

Out[10]:

| | OrderID | Quantity | Price | |
|---|---|---|---|---|
| count | 120.000000 | 120.000000 | 120.000000 | |
| mean | 60.500000 | 2.500000 | 4778.166667 | 1 |
| std | 34.785054 | 1.173947 | 4610.362749 | 1 |
| min | 1.000000 | 1.000000 | 999.000000 | |
| 25% | 30.750000 | 1.000000 | 1499.000000 | |
| 50% | 60.500000 | 3.000000 | 2499.000000 | |
| 75% | 90.250000 | 4.000000 | 4999.000000 | 1 |
| max | 120.000000 | 4.000000 | 12999.000000 | 5 |

In [11]:
```
total_sales = df["Total"].sum()
avg_order_value = df["Total"].mean()
print("Total sales:", round(total_sales, 2))
print("Average order value:", round(avg_order_value, 2))
```

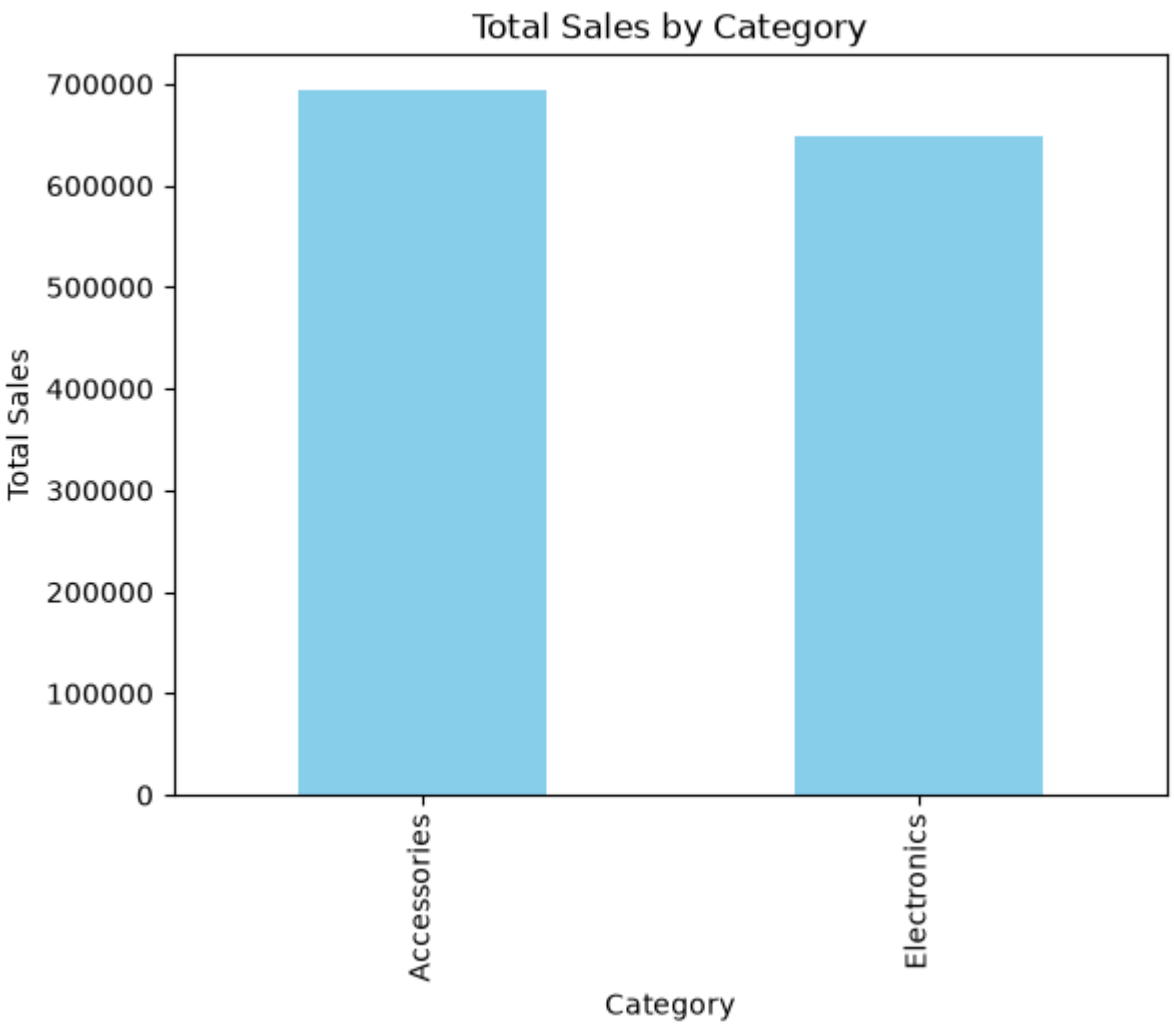
```
Total sales: 1341700.0
Average order value: 11180.83
```

In [12]:
```
sales_by_category = df.groupby("Category")["Total"].sum()
print("Total sales by category:")
print(sales_by_category)
print()
print("Number of orders by category:")
print(df["Category"].value_counts())
```

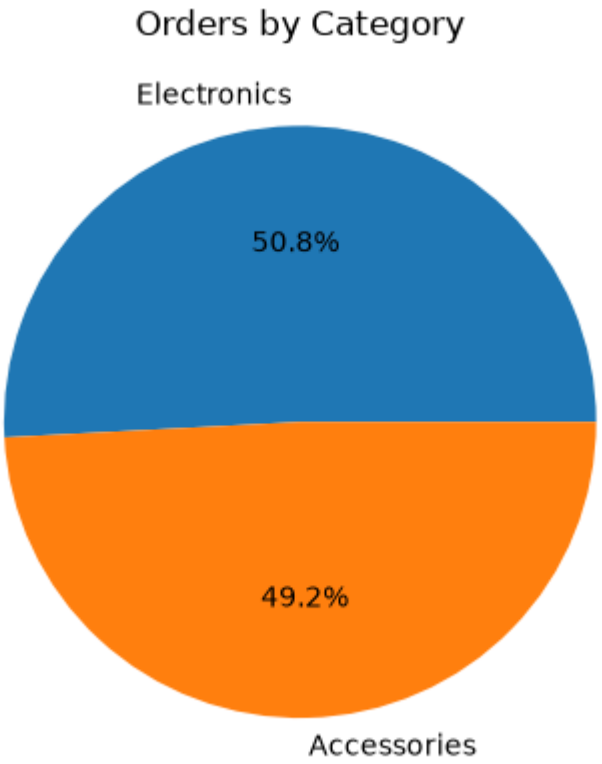
```
Total sales by category:
Category
Accessories    693855.0
Electronics    647845.0
Name: Total, dtype: float64

Number of orders by category:
Category
Electronics    61
Accessories    59
Name: count, dtype: int64
```

In [13]:
```
sales_by_category.plot(kind="bar", color="skyblue")
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.show()
```




In [14]:
```
df["Category"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Orders by Category")
plt.ylabel("")
plt.show()
```




In [15]:
```
df.boxplot(column="Price", by="Category")
plt.title("Price Distribution by Category")
plt.suptitle("")
plt.xlabel("Category")
plt.ylabel("Price")
plt.show()
```

